

Rohan Fozdar

rfozdar@knox.edu · +1 (646) 749-0816 · rohanfozdar.com

SUMMARY

Mathematics & Data Science student at Knox College, building AI systems to solve real-world problems. Experience spans quantitative modeling, LLM-based applications, and time-series forecasting, including an ERCOT price spike predictor and a RAG-driven job search engine. Background includes internships in AI automation and quantitative analytics, along with 2+ years as a Mathematics Teaching Assistant. Interested in AI/ML engineering and implementation roles.

TECHNICAL SKILLS

AI / ML	TensorFlow, PyTorch, HuggingFace, LangChain, FAISS, TTM
Languages	Python, JavaScript, R, SQL, C++, Java, Matlab
Quant / Data	NumPy, Pandas, Matplotlib, Excel, PowerBI, Tableau, Snowflake, FastAPI

EDUCATION

B.S. Mathematics & Data Science

Anticipated June 2027

Knox College, Galesburg, IL · GPA: 3.89

- › Dean's List (9 semesters) · Annual Carr Math Exam Winner · Knox Next Genius Scholar
- › Coursework: Probability Theory, Monte Carlo Simulations, Stochastic Processes, Options Theory & Portfolio Optimization, Applied Data Structures, Regression, Time Series (GARCH), Risk Management, Neural Networks & Deep Learning, NLP, Algorithm Design, Database Systems

PROJECTS rohanfozdar.com

ERCOT Electricity Price Forecaster

2026 – Present

- › Built an ERCOT electricity spot price forecaster using XGBoost trained on 9 data sources across 2016–2024, achieving 58.8% spike rate accuracy vs. the day-ahead market's 28.9%, detecting grid stress events up to 144 hours ahead.
- › Extending the architecture with IBM Research's Tiny Time Mixers (TTM), combining zero-shot temporal reasoning with domain-specific sentiment features to improve extreme price variation prediction.

Promotion Gauger

2026 – 2026

<https://promotion-gauger-gpizk83vr88fbfu8ijlaho.streamlit.app/>

- › Built a promotion sentiment tool that ingests live data from Reddit RSS and Google News, scores each source across three axes (price, brand, urgency) using a fine-tuned RoBERTa transformer.
- › Engineered a pipeline that performs text cleaning, relevance filtering, and validated against real-world known-outcome campaigns.

AutoSearch — AI Job Search Platform

2026 – 2026

job-finder-aa5m8lsoa-rohanfozdars-projects.vercel.app

- › Shipped an end-to-end job search platform with a REST API ingestion pipeline, RAG-style semantic scoring using FAISS vector search and sentence-transformers, and an ATS scoring microservice to surface best-fit roles.
- › Engineered an LLM-based ranking engine using PyMuPDF resume parsing, overlapping text chunking, cosine similarity matching against a FAISS vector store, and NLP-based job classification across 50+ listings per query.

Binomial Options Pricer

2025 – 2025

rohanfozdar.github.io/options-pricer/

- › Developed and deployed an options pricing tool via binomial branching with live YFinance data, providing valuations across all possible strikes and expirations with greater accuracy than closed-form alternatives.
- › Integrated automated risk-free rate calculations by pulling US Treasury yields based on expiration date; implemented a 3D IV surface visualisation with a market sentiment explainer module.

PROFESSIONAL EXPERIENCE

Quantitative Analytics Intern

Nov 2025 – Jan 2026

Monarch Network Capital
Mumbai, India

- › Designed a momentum trading strategy on NSE ETFs using time trend derivatives. Backtest achieved 45% returns & 3.43 Sharpe (BankNifty over 1 year).
- › Formulated a moving average strategy for NSE ETFs using CMP against 5/10/20/50/200 MA to indicate daily trends. Backtested 25 years of daily data achieving 3,100% returns & 2.86 Sharpe (Nifty over 25 years).

Mathematics Teaching Assistant

Mar 2024 – Present

Knox College Department of Mathematics
Galesburg, IL

- › Developed class material and led recitation sessions for Calculus I–III, Discrete Math, Linear Algebra, Foundations of Analysis, Methods of Proof, Differential Equations, Mathematical Statistics, and Financial Math.
- › Graded homework and exams; co-hosted department-wide tutoring sessions with fellow TAs.

AI Intern

Jun 2023 – Aug 2023

MeasureOne
Ahmedabad, India

- › Fine-tuned early ChatGPT models with prompt–completion pairs to generate specialised parser code. Designed a preprocessing pipeline that normalised heterogeneous transcript formats.
- › Reduced manual engineering workload by 25%.

LEADERSHIP & ORGANISATIONS

- › Knox Consulting Club — President
- › Honor Board — Co-chair
- › Quantitative Investment Society